



An efficient multiparameter acoustic anisotropic full-waveform inversion depending on parameterization

Youngjae Shin¹ · Ju-Won Oh² · Dong-Joo Min^{3,4}

Received: 10 September 2020 / Accepted: 13 May 2021 / Published online: 6 July 2021
© Institute of Geophysics, Polish Academy of Sciences & Polish Academy of Sciences 2021

Abstract

The pressure-based acoustic approximation of the elastic wave equations in anisotropic media has advantages corresponding to computational efficiency and numerical stability. However, the numerical scattering potentials from the anisotropic parameter perturbations for the pressure wavefield are not consistent with the elastic scattering theory. In multiparameter full-waveform inversion (FWI), choosing a suitable parameterization, considering the acquisition parameters (e.g., the offset-to-depth ratio and frequency band) and the accuracy of the anisotropy information in the background initial velocity model, is an important component to a successful anisotropic parameter estimation, because the parameterization determines the trade-off between inverted model parameters and their resolution power. However, because it is difficult to perform multiparameter FWI with various types of parameterization using the pressure-based acoustic wave equation, inaccurate scattered wavefields cause the gradient direction to lose its unique properties with respect to each model parameter. To overcome these issues, we adopt the combination of pressure- and vector-based acoustic wave equations converted vector virtual sources, which preserves the computational efficiency and the angular dependency of the partial derivative wavefields in elastic media. With the proposed method, we generate the numerical PP scattering patterns for various parameterizations, which are consistent with the elastic scattering theory. Through the numerical tests using the synthetic anisotropic Marmousi-II models and a real ocean bottom cable dataset from the North Sea, we conduct acoustic FWI with three different parameterizations using the proposed method and verify that the modified scattering patterns accurately reflect the characteristics of the anisotropic parameter perturbations.

Keywords Acoustic · Anisotropy · Full-waveform inversion · Parameterization · Ocean bottom cable

Introduction

To retrieve physical properties and structures of subsurface media, full-waveform inversion (FWI) has been used as a nonlinear optimization technique for obtaining high-resolution images (Lailly 1983; Tarantola 1984; Virieux and Operto 2009). Although various physical properties including the density and elastic moduli are required to describe wave propagation in complex anisotropic media, it is important to set assumptions suitable for certain study areas to mitigate ill-posedness of the inverse problem by reducing the number of variables. In this context, for the commonly encountered anisotropic medium, i.e., the vertical transverse isotropic (VTI) medium, which is symmetric along the vertical axis of the material, the acoustic-approximated FWI methods based on the pseudo-acoustic wave equations have been designed as an alternative to elastic FWI (Alkhalifah 2000; Zhou et al. 2006; Xu and Zhou 2014).

Communicated by Prof. Sanyi Yuan (ASSOCIATE EDITOR) / Prof. Michał Malinowski (CO-EDITOR-IN-CHIEF).

✉ Ju-Won Oh
juwon24@jbnu.ac.kr

- ¹ Mineral Resources Research Division, Korea Institute of Geoscience and Mineral Resources, 124, Gwahak-ro, Yuseong-gu, Daejeon 34132, Republic of Korea
- ² Department of Mineral Resources and Energy Engineering, Jeonbuk National University, 567, Baekje-daero, Deokjin-gu, Jeonju-si, Jeollabuk-do 54896, Republic of Korea
- ³ Department of Energy Systems Engineering, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea
- ⁴ Research Institute of Energy and Resources, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea

Since the pseudo-acoustic wave equation was first proposed to approximately simulate wave propagation in VTI media, various equations have been presented to improve the accuracy and efficiency of pseudo-acoustic modeling. The early pseudo-acoustic wave equations consist of coupled second-order differential equations (Zhou et al. 2006; Duveneck et al. 2008; Gholami et al. 2013a), which will be referred to as “the coupled equations” throughout the paper. The coupled equations correctly describe the amplitudes (the geometrical spreading) of waves propagating in VTI media, but shear wave artifacts are generated. To avoid the problem of generating shear wave artifacts, the methods of decomposing the original pseudo-acoustic wave equation were proposed (Xu and Zhou 2014; Xu et al. 2015; Wu et al. 2018), which will be referred to as “the decomposed equation” throughout the paper. The decomposed methods have another advantage as they are computationally more efficient than the coupled method.

Although the decomposed equation could describe waves propagating in VTI media, they have a limitation when used in multiparameter FWI. Numerical scattering patterns due to the anisotropic parameter perturbations computed by the decomposed equation deviate from the theoretical scattering patterns (Shin et al. 2019). To compensate for this limitation, Shin et al. (2019) proposed a strategy to use different modeling engines for the forward and inverse problems. In other words, they calculate the gradient based on the coupled equations with the vector wavefields converted from pressure, which can reconstruct theoretically correct radiation patterns.

One of the factors for the success of multiparameter FWI is to choose an appropriate parameterization considering the acquisition parameters (offset-to-depth ratio) and the accuracy of anisotropy information in the background initial velocity model. To properly simulate waves propagating in acoustic VTI media using the decomposed equations, three parameters are generally required. We can use various combinations of the parameters for the forward and inverse problems. While forward modeling results are not dependent on the parameterization, the inverse problem is significantly affected by the parameterization, because it determines inversion coverage of each parameter and the trade-off among model parameters. Alkhalifah (2016) proposed to use the horizontal P-wave velocity v_{ph} , the anelliptic parameter η (Alkhalifah and Tsvankin 1995), and Thomsen’s parameter (Thomsen 1986) ε in the acoustic-approximated FWI for VTI media because the horizontal velocity can be updated by the long-wavelength components and suffers relatively less from the trade-off between parameters (also supported by Kamath et al. (2017), and Oh and Alkhalifah (2018)), and η has little influence in the case of the conventional surface acquisition geometry (Oh et al. 2020). Gholami et al. (2013a; 2013b) analyzed the scattering patterns with respect

to model parameters according to two types of parameterization. The first type of parameterization consisting of one velocity and two anisotropic parameters produces a high-resolution velocity model when the initial models of the anisotropic parameters are accurate. The second type of parameterization consisting of two velocity parameters and one anisotropic parameter is suitable for updating the vertical velocity with near-offset data and the horizontal velocity with far-offset data by reducing the trade-off between model parameters.

In this study, extending the work by Shin et al. (2019), we seek an accurate and efficient strategy for acoustic-approximated multiparameter VTI FWI using the decomposed equation in terms of parameterization. We first review the forward and inverse problem based on the acoustic approximation in 2D VTI media using the decomposed equation with the converted wavefields from pressure. Then, we analyze the scattering patterns with respect to model parameters for three different parameterizations. Finally, we conduct FWI using the three parameterizations for synthetic data generated for the Marmousi-II model and a real ocean bottom cable (OBC) dataset acquired in the Volve oil field of the North Sea.

Multiparameter anisotropic full-waveform inversion

To efficiently and accurately conduct anisotropic FWI for 2D VTI media, we introduce the combination of the decomposed equation for the forward problem and the coupled equation for the inversion problem (Shin et al. 2019).

The forward problem using the decomposed equation

The wave propagation in acoustic VTI media can be described by various techniques, but we focus on the coupled equation and the decomposed equation in this paper. The coupled equations consist of multicomponent acoustic wave equations, which are similar to the elastic wave equations excluding the shear strain terms. Thus, the coupled equations generate the PP waves that match well with those obtained by the elastic wave equations in terms of amplitude and phase, but its calculation speed is slower than that of the decomposed equation, and they suffer from shear wave artifacts. The decomposed equation, which consists of the pressure wavefield with the differential and scalar operators, has an advantage of computational efficiency without producing shear wave artifacts (Xu and Zhou 2014). In this study, we choose the elliptic decomposition of the original pseudo-differential equation (Xu et al. 2015), which can

reduce numerical errors for strongly anisotropic media as follows:

$$\frac{\partial^2 P}{\partial t^2} = S_e(M_e P) + f_p, \quad (1)$$

$$S_e = \frac{1}{2} \left(1 + \sqrt{1 - \frac{8(\varepsilon - \delta)n_x^2 n_z^2}{((1 + 2\varepsilon)n_x^2 + n_z^2)^2}} \right), \quad (2)$$

$$M_e = \left(v_{pv}^2 (1 + 2\varepsilon) \frac{\partial^2}{\partial x^2} + v_{pv}^2 \frac{\partial^2}{\partial z^2} \right), \quad (3)$$

where P is the pressure wavefield, f_p is the pressure source, v_{pv} is the vertical P-wave velocity, ε and δ are Thomsen's parameters (Thomsen 1986), and t is time. M_e and S_e are the elliptic differential and scalar operators, respectively. The terms, n_x and n_z , are the unit vectors for the phase direction in the horizontal and vertical axes, respectively. The unit vectors in the scalar operator can be replaced by the normalized gradient of the pressure wavefield to solve the wave equation in the time–space domain, which is based on the fact that the unit vectors physically mean the phase direction of wavefront (Xu and Zhou 2014).

The inverse problem using the coupled equation

Because the decomposed equation is given by an elliptical background, the numerical scattering potentials due to the model parameter perturbations have isotropic scattering patterns regardless of parameter (Shin et al. 2019). These incorrect scattering patterns can lead to inaccurate gradient directions and suffering from unwanted trade-offs between model parameters. In contrast, the PP scattering patterns obtained by the coupled equations conform to those of the elastic scattering theory for an acoustic model (Gholami et al. 2013a). Therefore, the combination of the forward problem using the decomposed equation and the inversion problem using the coupled equation can provide a reasonable strategy for acoustic anisotropic multiparameter FWI as Shin et al. (2019) proposed.

The l_2 -norm-based objective function can be defined as follows:

$$E(m) = \frac{1}{2} \sum_s \sum_r \sum_t [W(r)P^m(\mathbf{x}, s, t, \mathbf{m}) - P^o(s, r, t)]^2, \quad (4)$$

where $\mathbf{x}$ represents the spatial coordinates for the horizontal and vertical directions; s and r are the source and the receiver locations, respectively; P^m indicates the modeled pressure wavefield at each nodal point; and P^o is the observed pressure wavefield at the receivers. The function $W(r)$ denotes the mapping function to project the modeled

wavefields to the receiver positions. $\mathbf{m}$ denotes the physical properties including the anisotropic parameters. The modeled pressure wavefield (P^m) is calculated by the decomposed wave equation in Eq. (1).

For a local minimization of the objective function, the gradient direction at the i^{th} nodal point can be obtained by taking a partial derivative of the objective function with respect to the model parameter (m_i) and by computing the zero-lag cross-correlation between the partial derivative and the residual wavefields as follows:

$$\begin{aligned} \frac{\partial E(\mathbf{m})}{\partial m_i} &= \sum_s \sum_r \sum_t \left[W(r) \frac{\partial P^m(\mathbf{x}, s, t, \mathbf{m})}{\partial m_i} [W(r)P^m(\mathbf{x}, s, t, \mathbf{m}) - P^o(s, r, t)] \right]. \end{aligned} \quad (5)$$

The scalar partial derivative wavefields $\frac{\partial P^m}{\partial m_i}$ can be obtained by taking a derivative of the wave equation with respect to the i th model parameter at each source and time step as follows:

$$\frac{\partial^2}{\partial t^2} \left(\frac{\partial P^m(\mathbf{x}, \mathbf{m})}{\partial m_i} \right) = S_e(\mathbf{m}) \left(M_e(\mathbf{m}) \frac{\partial P^m(\mathbf{x}, \mathbf{m})}{\partial m_i} \right) + f_{p, m_i}^v(\mathbf{x}, \mathbf{m}), \quad (6)$$

$$f_{p, m_i}^v(\mathbf{x}, \mathbf{m}) = \left(\frac{\partial M_e(\mathbf{m})}{\partial m_i} S_e(\mathbf{x}, \mathbf{m}) + M_e(\mathbf{m}) \frac{\partial S_e(\mathbf{x}, \mathbf{m})}{\partial m_i} \right) P^m(\mathbf{x}, \mathbf{m}), \quad (7)$$

where f_{p, m_i}^v is the virtual source (Pratt et al. 1998), which acts as a source term for the partial derivative wavefields. To accelerate the calculation of the gradient direction, the adjoint-state method is used (Lailly 1983; Pratt et al. 1998; Plessix 2006). The gradient direction at the i th nodal point obtained using the decomposed equation based on the adjoint-state method can be expressed by

$$\frac{\partial E(\mathbf{m})}{\partial m_i} = \sum_s \sum_t \left[f_{p, m_i}^v(i, s, t, \mathbf{m}) \cdot P^b(i, s, T - t, \mathbf{m}) \right] \quad (8)$$

where P^b denotes the back-propagated pressure wavefields obtained by propagating the residual wavefields to the whole model domain backward from the recording time T to 0. Because the scalar virtual source cannot produce directional scattering patterns depending on model parameters, PP waves of the partial derivative wavefields due to the anisotropic parameter perturbations deviate from the elastic scattering theory.

To overcome this limitation, Shin et al. (2019) proposed computing the vector virtual sources based on the coupled equations to reconstruct the radiation patterns of anisotropic parameters. The coupled equations from the elastic wave equations for 2D VTI media can be derived by setting the S-wave velocity to be 0 at each source and time step as follows:

$$a_x(\mathbf{x}, \mathbf{m}) = \frac{1}{\partial x} \left(C_{11} \frac{\partial u_x(\mathbf{x}, \mathbf{m})}{\partial x} + C_{13} \frac{\partial u_z(\mathbf{x}, \mathbf{m})}{\partial z} \right) + f_{a_x}(\mathbf{x}), \quad (9)$$

$$a_z(\mathbf{x}, \mathbf{m}) = \frac{1}{\partial z} \left(C_{13} \frac{\partial u_x(\mathbf{x}, \mathbf{m})}{\partial x} + C_{33} \frac{\partial u_z(\mathbf{x}, \mathbf{m})}{\partial z} \right) + f_{a_z}(\mathbf{x}), \quad (10)$$

where $\mathbf{a} = (a_x, a_z)$ denotes the particle accelerations, $\mathbf{u} = (u_x, u_z)$ means the particle displacements, and $\mathbf{f}_a = (f_{a_x}, f_{a_z})$ represents the acceleration sources. The stiffness tensor C_{ij} defines the relationship between stress and strain. The virtual sources by the coupled equations can be obtained by taking a partial derivative of Eqs. (9) and (10) with respect to the model parameters at each source and time step as follows:

$$\frac{\partial \mathbf{a}(\mathbf{x}, \mathbf{m})}{\partial m_i} = \begin{pmatrix} C_{11} \frac{\partial^2}{\partial x^2} & C_{13} \frac{\partial^2}{\partial z \partial x} \\ C_{13} \frac{\partial^2}{\partial x \partial z} & C_{33} \frac{\partial^2}{\partial z^2} \end{pmatrix} \begin{pmatrix} \frac{\partial u_x(\mathbf{x}, \mathbf{m})}{\partial m_i} \\ \frac{\partial u_z(\mathbf{x}, \mathbf{m})}{\partial m_i} \end{pmatrix} + \mathbf{f}_{a, m_i}^v(\mathbf{x}) \quad (11)$$

$$\mathbf{f}_{a, m_i}^v(\mathbf{x}, \mathbf{m}) = \frac{\partial}{\partial m_i} \begin{pmatrix} C_{11} \frac{\partial^2}{\partial x^2} & C_{13} \frac{\partial^2}{\partial z \partial x} \\ C_{13} \frac{\partial^2}{\partial x \partial z} & C_{33} \frac{\partial^2}{\partial z^2} \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix}, \quad (12)$$

where $\mathbf{f}_{a, m_i}^v$ is the virtual source in terms of particle acceleration, which includes multicomponents and spatial derivatives to determine the radiation pattern. The particle displacement $\mathbf{u}$ is required to calculate the vector virtual sources using Eq. (12) from the acoustic VTI wave equation. By the first-order form of the decomposed wave equation, the pseudo-particle accelerations, which can be converted into particle displacements, are approximated by the pressure wavefields as follows:

$$\mathbf{a}(\mathbf{x}, \mathbf{m}) = \nabla P^m(\mathbf{x}, \mathbf{m}) \quad (13)$$

and

$$\begin{aligned} & \frac{\partial^2 P^m(\mathbf{x}, \mathbf{m})}{\partial t^2} \\ &= S_e(\mathbf{x}, \mathbf{m}) \left(v_{pv}^2 (1 + 2\varepsilon) \frac{\partial a_x(\mathbf{x}, \mathbf{m})}{\partial x} + v_{pv}^2 \frac{\partial a_z(\mathbf{x}, \mathbf{m})}{\partial z} \right) \\ &+ f_p(\mathbf{x}, \mathbf{m}), \end{aligned} \quad (14)$$

Finally, the gradient direction obtained using the coupled equation based on the adjoint-state method can be calculated

$$\frac{\partial E(\mathbf{m})}{\partial m_i} = \sum_s \sum_t \left[\mathbf{f}_{a, m_i}^v(\mathbf{x}, s, t, \mathbf{m})^T \cdot \mathbf{a}^B(\mathbf{x}, s, T - t, \mathbf{m}) \right] \quad (15)$$

where $\mathbf{a}^B$ denotes the back-propagated particle acceleration using the adjoint formula of Eq. (14). With the vector-based gradient, we can update the anisotropic parameters in the correct gradient direction preserving the computational efficiency of the forward modeling scheme. In addition, the

virtual source of the anisotropic parameters depending on parameterization can be easily calculated considering the relation between elastic constants (C_{ij}) in Eq. (12) and a parameter set we choose for FWI. The scattering patterns by parameterization will be described in the following section.

Scattering pattern analysis of VTI FWI

For acoustic VTI media, three model parameters are required to describe the wave equations. The analysis of the scattering patterns for different incident angles according to parameterization has been shown in the previous studies using the coupled equation (Gholami et al. 2013a; Plessix and Cao 2011). However, when the decomposed equation is used for FWI, it was difficult to obtain the accurate radiation patterns because of the isotropic pressure virtual source. Thus, we show the scattering patterns of partial derivative wavefields described by using the vector wavefields converted from pressure for three kinds of parameterizations.

Scattering potentials for different parameterizations

In multiparameter anisotropic FWI, it is important to analyze the scattering patterns of the partial derivative wavefields with respect to model parameters to investigate the trade-off and the potential resolution power of the inversion. Because the shape of the radiation patterns varies according to parameterization, we can choose an optimal parameterization considering the acquisition geometry and background models (Gholami et al. 2013b).

The partial derivative wavefields are computed by taking the virtual source as a source term, and the type of virtual source determines the variation of the scattering patterns over incidence angles (Oh and Alkhalifah 2016). We compare the radiation patterns of three parameterizations: $[v_{pv}, \varepsilon, \delta]$, $[v_{ph}, \varepsilon, \delta]$, and $[v_{pv}, v_{ph}, \delta]$. The vector virtual sources in Eq. (12) using the first parameterization $[v_{pv}, \varepsilon, \delta]$ can be expressed as:

$$\mathbf{f}_{a, v_{pv}}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 2v_{pv}(1 + 2\varepsilon) \frac{\partial^2}{\partial x^2} & 2v_{pv} \sqrt{1 + 2\delta} \frac{\partial^2}{\partial z \partial x} \\ 2v_{pv} \sqrt{1 + 2\delta} \frac{\partial^2}{\partial x \partial z} & 2v_{pv} \frac{\partial^2}{\partial z^2} \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix} \quad (16)$$

$$\mathbf{f}_{a, \varepsilon}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 2v_{pv}^2 \frac{\partial^2}{\partial x^2} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix} \quad (17)$$

$$\mathbf{f}_{a, \delta}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 0 & \frac{v_{pv}^2}{\sqrt{1 + 2\delta}} \frac{\partial^2}{\partial z \partial x} \\ \frac{v_{pv}^2}{\sqrt{1 + 2\delta}} \frac{\partial^2}{\partial x \partial z} & 0 \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix}. \quad (18)$$

Equations (16) to (18) show that the virtual sources, which will determine the radiation patterns of partial derivative wavefields, are dependent on the spatial derivatives (Oh and Alkhalifah 2016). Accordingly, the virtual sources consisting of different spatial derivatives will yield different gradient directions. Figure 1a shows the numerical and theoretical PP scattering patterns of the partial derivative wavefields for different incident angles using the scalar and vector virtual sources for the $[v_{pv}, \epsilon, \delta]$ parameterization.

The radiation patterns of v_{pv} using both the scalar and vector virtual sources are isotropic at every incident angles, which are consistent with the analytic solution. This isotropic pattern over all incident angle results in high

resolution of inverted model parameters including both high and low wavenumbers, while the trade-off with other model parameters is inevitable.

As opposed to the scattering pattern of v_{pv} , the shape of scattered wavefields for the anisotropic parameters using the scalar virtual source is distinct from the theoretical scattering pattern (Fig. 1a). The dominant offset ranges for FWI in surface acquisition with respect to the model parameters are similar to the analytic result, but the directional information of the scattered waves is not included, which can lead to incorrect gradient directions for the anisotropic parameters. The scattered wavefields due to the perturbation of ϵ using the vector virtual sources propagate horizontally at narrow angles and vertically at wide angles, which also agrees with the theoretical scattering pattern. Because the amplitude of scattered waves increases at wide incident angle, ϵ is mainly updated by the long-wavelength components for surface acquisition. The scattered wavefields of δ using the vector virtual sources propagate horizontally at narrow angles and vertically at wide angles, which also agrees with the theoretical pattern of δ . Under the geometry of surface acquisition, it is not desirable to update δ due to the intrinsic limited coverage. Therefore, the $[v_{pv}, \epsilon, \delta]$ parameterization is suitable to obtain high-resolution v_{pv} with near-offset reflection data assuming that the anisotropic parameters are well constructed with far-offset data for surface acquisition.

The vector virtual sources by the $[v_{ph}, \epsilon, \delta]$ parameterization can be calculated similar to Eqs. (16), (17) and (18) as follows:

$$\mathbf{f}_{a,v_{ph}}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 2v_{ph} \frac{\partial^2}{\partial x^2} & 2v_{ph} \frac{\sqrt{1+2\delta}}{(1+2\epsilon)} \frac{\partial^2}{\partial z \partial x} \\ 2v_{ph} \frac{\sqrt{1+2\delta}}{(1+2\epsilon)} \frac{\partial^2}{\partial x \partial z} & 2v_{ph} \frac{1}{(1+2\epsilon)} \frac{\partial^2}{\partial z^2} \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix} \quad (19)$$

$$\mathbf{f}_{a,\epsilon}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 0 & -2v_{ph}^2 \frac{\sqrt{1+2\delta}}{(1+2\epsilon)^2} \frac{\partial^2}{\partial z \partial x} \\ -2v_{ph}^2 \frac{\sqrt{1+2\delta}}{(1+2\epsilon)^2} \frac{\partial^2}{\partial x \partial z} & -2v_{ph}^2 \frac{1}{(1+2\epsilon)^2} \frac{\partial^2}{\partial z^2} \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix} \quad (20)$$

$$\mathbf{f}_{a,\delta}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 0 & \frac{v_{ph}^2}{\sqrt{1+2\delta}(1+2\epsilon)} \frac{\partial^2}{\partial z \partial x} \\ \frac{v_{ph}^2}{\sqrt{1+2\delta}(1+2\epsilon)} \frac{\partial^2}{\partial x \partial z} & 0 \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix}. \quad (21)$$

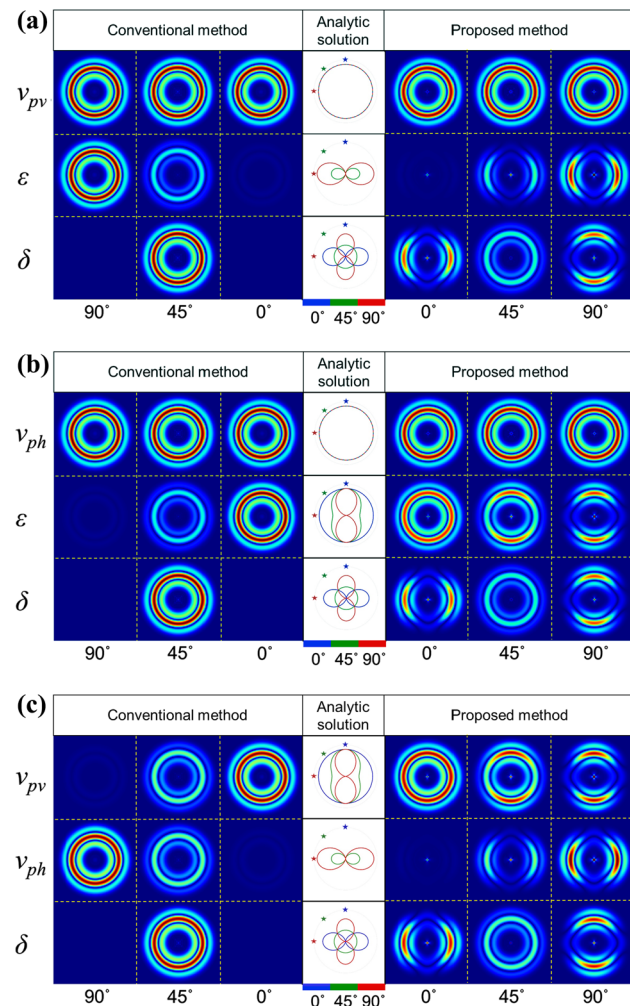


Fig. 1 Numerical (left and right) and theoretical (center) PP scattering patterns of the partial derivative wavefields obtained at incident angles of 0°, 45°, and 90° using the pressure virtual source (left) and converted vector virtual sources (right) for **a** the $[v_{pv}, \epsilon, \delta]$, **b** $[v_{ph}, \epsilon, \delta]$, and **c** $[v_{pv}, v_{ph}, \delta]$ parameterizations. Colors of radiation patterns in analytic solutions (center) represent the incident angles: 0° (blue), 45° (green), and 90° (red). The colored stars in analytic solutions (center) denote the source locations at each incident angle

Figure 1b represents the numerical and theoretical PP scattering patterns of the partial derivative wavefields at different incident angles using the scalar and vector virtual sources for the $[v_{ph}, \epsilon, \delta]$ parameterization. The scattered wavefields due to the perturbation of v_{ph} have an isotropic pattern using both the scalar and vector virtual sources, which cover a full range of incident angles. Unlike the $[v_{pv}, \epsilon, \delta]$ parameterization, in surface acquisition, ϵ combined with v_{ph} is mainly affected by the data at narrow incident

angles, and the vertically spreading wavefields appear at intermediate and wide incident angles in the theoretical and numerical scattering patterns using the vector virtual sources, which are not described in the scattering pattern using the scalar virtual source. The long-wavelength component of v_{ph} can be recovered by reducing the trade-off with the other parameters. For narrow incident angles, both v_{ph} and ε are updated mainly by the reflection data.

The anelliptic parameter η can be a substitute for δ , which generates the different scattering patterns of the anisotropic parameters. When the $[v_{ph}, \varepsilon, \eta]$ parameterization is used, it is suitable to invert both ε and η because the trade-off between ε and η is relatively small (Alkhalifah 2016). However, it is difficult to invert either δ or η for surface acquisition with limited offsets; therefore, we focus on the updates of velocity and ε without inverting the third parameter in this paper. For this purpose, it is advantageous to choose the $[v_{ph}, \varepsilon, \delta]$ parameterization because the coverage of scattering pattern for ε combined with δ is wider than that combined with η . If η has little influence on the anisotropy of a certain anisotropic area and with limited offsets, the $[v_{ph}, \varepsilon, \eta]$ parameterization can unnecessarily narrow the scattering pattern of ε . Therefore, the $[v_{ph}, \varepsilon, \delta]$ parameterization is suitable to recover long wavelengths of v_{ph} independently and estimate ε with reflection data for surface acquisition without updating δ .

The vector virtual sources by the $[v_{pv}, v_{ph}, \delta]$ parameterization can be represented by

$$\mathbf{f}_{a,v_{pv}}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 0 & 2v_{pv}\sqrt{1+2\delta}\frac{\partial^2}{\partial z\partial x} \\ 2v_{pv}\sqrt{1+2\delta}\frac{\partial^2}{\partial x\partial z} & 2v_{pv}\frac{\partial^2}{\partial z^2} \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix}, \quad (22)$$

$$\mathbf{f}_{a,v_{ph}}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 2v_{ph}\frac{\partial^2}{\partial x^2} & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix}, \quad (23)$$

$$\mathbf{f}_{a,\delta}^v(\mathbf{x}, \mathbf{m}) = \begin{pmatrix} 0 & \frac{v_{pv}^2}{\sqrt{1+2\delta}}\frac{\partial^2}{\partial z\partial x} \\ \frac{v_{pv}^2}{\sqrt{1+2\delta}}\frac{\partial}{\partial x\partial z} & 0 \end{pmatrix} \begin{pmatrix} u_x(\mathbf{x}, \mathbf{m}) \\ u_z(\mathbf{x}, \mathbf{m}) \end{pmatrix}. \quad (24)$$

Figure 1c shows the numerical and theoretical PP scattering patterns of the partial derivative wavefields at different incident angles using the scalar and vector virtual sources by $[v_{pv}, v_{ph}, \delta]$ parameterization. This parameterization is structurally different from the previous two parameterizations because it consists of two velocities (v_{pv} and v_{ph}) and one anisotropic parameter δ . The limited coverage of the P-wave velocity in Fig. 1c can degrade the resolution of FWI results compared to the P-wave velocity by the previous two parameterizations (Kamath et al. 2017). For surface acquisition, while v_{pv} is mainly updated by the vertical wavepaths, v_{ph} is recovered from wide opening angle components. Thus,

the trade-off between two velocities can be minimized due to the small overlapping part. However, using the conventional method, the unwanted trade-off between v_{pv} and v_{ph} occurs at an incident angle of 45 degree, where the scattered waves for v_{pv} and v_{ph} have different propagation directions theoretically.

All of these results demonstrate that the scattering patterns with respect to model parameters obtained with the vector virtual sources, which are computed by deriving the vector wavefields from the decomposed equation, agree well with the theoretical radiation patterns for the three parameterizations. Therefore, we can choose an optimal parameterization suitable for survey environments while maintaining the computational speed-up of the decomposed equation.

Numerical examples

We compare FWI results for different parameterizations by conducting FWI for the anisotropic Marmousi-II models in surface acquisition. We also apply FWI with different parameterizations to the field data, i.e., the OBC dataset acquired from the Volve oil field of the North Sea.

The Marmousi-II model

To verify our acoustic multiparameter FWI strategy for the different parameterizations, we use the synthetic anisotropic Marmousi-II models shown in Fig. 2 (Alkhalifah 1997). The structure of ε was designed differently from that of v_{pv} specifically at the bottom and left part to investigate the trade-off between them. Because the surface seismic data are relatively less affected by δ , we fixed δ at zero. The starting model is the smoothed version of true model as shown in Fig. 2c–d, which means that the long-wavelength components are well constructed. To remove unwanted reflections from the artificial boundaries at the left, right, and bottom of the model, we apply the CPML (convolution perfectly matched layer) boundary condition (Komatitsch and Martin 2007). A Ricker wavelet with a dominant frequency of 5 Hz is used, and 80 sources are located on the surface at an interval of 120 m. 470 receivers are also placed on the surface from 0.2 to 9.6 km with an interval of 20 m. We invert all the parameters simultaneously excluding δ with constant step lengths, which are empirically obtained.

Figure 3 shows the inverted v_{pv} and ε by the $[v_{pv}, \varepsilon, \delta]$, $[v_{ph}, \varepsilon, \delta]$ and $[v_{pv}, v_{ph}, \delta]$ parameterizations. With the $[v_{ph}, \varepsilon, \delta]$ parameterization, v_{pv} was extracted from the inverted v_{ph} and ε , and with the $[v_{pv}, v_{ph}, \delta]$ parameterization, ε is estimated from the relationship between v_{pv} and v_{ph} .

Figure 3a to c shows that v_{pv} is properly inverted by all three parameterizations. Among them, the $[v_{ph}, \varepsilon, \delta]$ parameterization provides the highest resolution inverted velocity

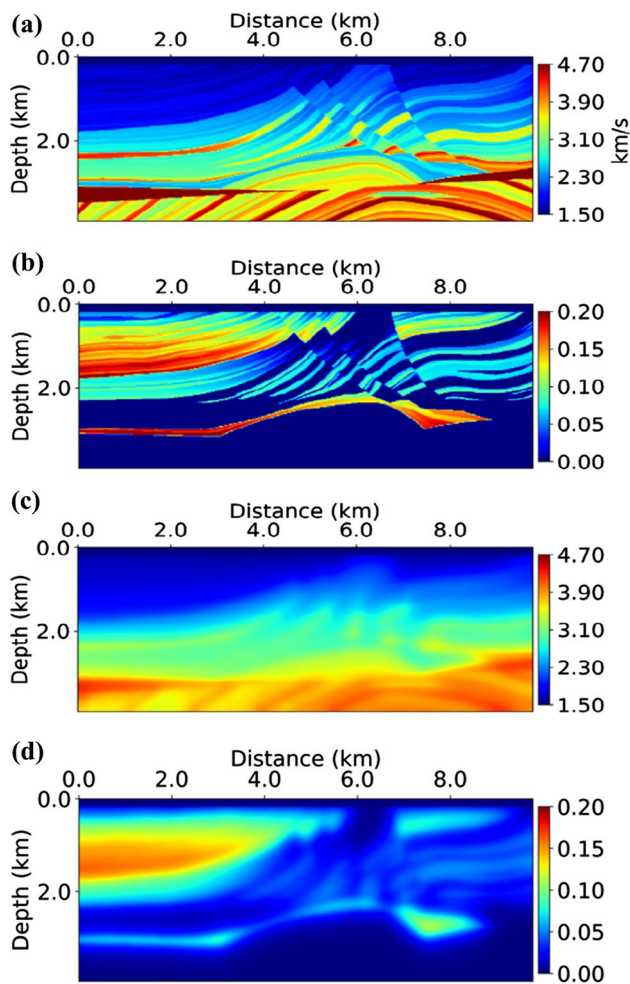


Fig. 2 **a, b** Anisotropic Marmousi-II models and **c, d** starting models for FWI: **a, c** v_{pv} and **b, d** ϵ . δ is fixed at zero

model because of the high wavenumber component of ϵ inverted with the tomographic model. In spite of the narrow range of scattering pattern of v_{pv} in the $[v_{pv}, v_{ph}, \delta]$ parameterization, it is also recovered well because the long-wavelength structure is already established in the initial model.

On the other hand, the inverted ϵ differs depending on parameterization. With the $[v_{pv}, \epsilon, \delta]$ parameterization, ϵ is not properly updated, because the scattered wavefields are mainly affected by data at wide reflection angles and the long-wavelength structure is already formed in the smoothed initial model (Fig. 3d). ϵ inverted by the $[v_{ph}, \epsilon, \delta]$ parameterization is close to the true model although the trade-off with v_{ph} occurs in the bottom part (Fig. 3e), which means that even a limited coverage from narrow reflection angles can produce reasonable FWI results if long-wavelength components are established by initial model. Therefore, the $[v_{ph}, \epsilon, \delta]$ parameterization is suitable for multiparameter FWI if the long-wavelength model of anisotropic parameter can be obtained by prior information such as traveltimes

tomography. ϵ updated by the $[v_{pv}, v_{ph}, \delta]$ parameterization includes some overestimated areas (Fig. 3f), because ϵ is calculated indirectly from the poorly updated v_{ph} due to the absence of near-offset data. Through these synthetic tests, we confirm that the corrected scattering patterns with vector virtual sources by parameterization contribute to FWI results that are consistent with the elastic scattering theory.

To examine the stability of our approach, FWI is carried out for noise-added data, while the other conditions are the same as in the previous experiment. The level of noise is set to 10% of the maximum amplitude of forward data. Figure 4 shows the inversion results of v_{pv} and ϵ by the three parameterizations when Gaussian noise is added to forward data. Although the inverted velocities have overall lower resolution than those with noise-free data, the differences for the parameterizations are clearly described in Fig. 4a to c. ϵ obtained by the $[v_{pv}, \epsilon, \delta]$ parameterization is highly contaminated by noise due to the trade-off with v_{pv} . With the $[v_{pv}, v_{ph}, \delta]$ parameterization, the overestimated parts in the inverted ϵ are observed regardless of noise.

OBC data of the Volve oil field of the North Sea

We also conduct acoustic anisotropic FWI with the three parameterizations for the OBC dataset collected from the Volve oil field in the North Sea. Among the 4C components, only pressure data are utilized, and we choose a single line among 16 ocean bottom cables for 2D FWI (cable number: 40195–1,252,860). The number of receivers is 238 whose average depth is about 80 m, and the pressure field from 110 airgun shots adjacent to the single cable is used for FWI. 3D reflection tomography models were also released along with the seismic data by Equinor and its former Volve license partners since October 2018. The 2D tomography background model in cable number 40195–1,252,860 is shown in Fig. 5. For processing of the field data, a bandpass filter (4–20 Hz) is applied and we convert 3D-to-2D pressure wavefields (Červený 2005) to account for the geometrical spreading effects. Figure 6 shows the smoothed tomography models used as an initial model for multiparameter FWI.

By the $[v_{pv}, \epsilon, \delta]$, $[v_{ph}, \epsilon, \delta]$, and $[v_{pv}, v_{ph}, \delta]$ parameterizations, we simultaneously update the first and second parameters of each parameterization with δ fixed. Figure 7 shows FWI results of v_{pv} and ϵ obtained by the three parameterizations. Because the long-wavelength structures of model parameters are established from the tomography models, the detailed structures at the top and bottom parts are described in the inverted P-wave velocities by the three parameterizations. In particular, the $[v_{ph}, \epsilon, \delta]$ parameterization produces higher resolution of inverted velocity than the other two parameterizations because the scattered energy of ϵ focuses on reflection data at near offsets. The sandstone reservoirs

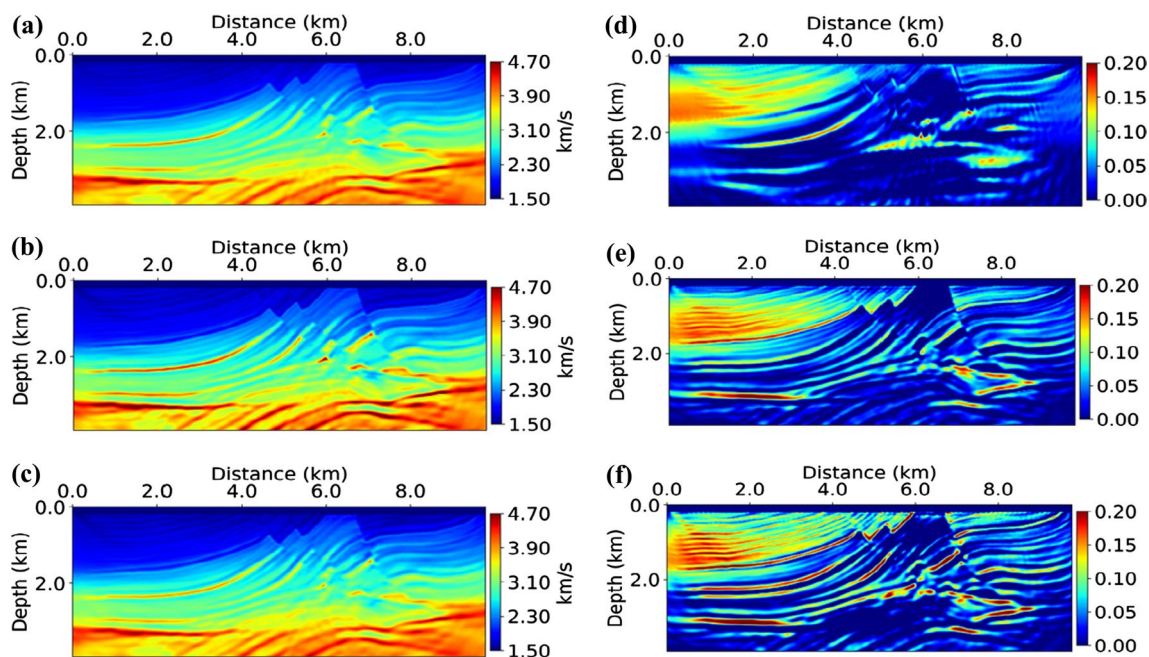


Fig. 3 FWI results of **a, b, c** v_{pv} and **d, e, f** ϵ obtained by **a, d** $[v_{pv}, \epsilon, \delta]$ parameterization, **b, e** $[v_{ph}, \epsilon, \delta]$ parameterization and **c, f** $[v_{pv}, v_{ph}, \delta]$ parameterization for the anisotropic Marmousi-II models

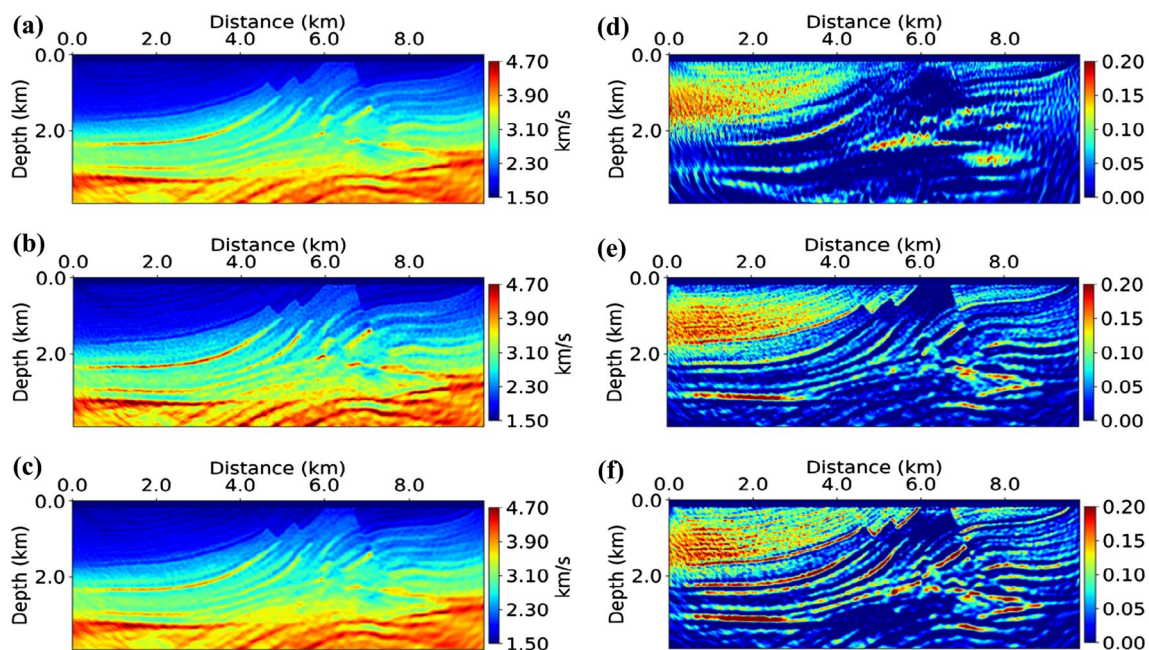


Fig. 4 FWI results from Gaussian noise-added data of **(a, b, c)** v_{pv} and **(d, e, f)** ϵ obtained by **a, d** $[v_{pv}, \epsilon, \delta]$ parameterization, **b, e** $[v_{ph}, \epsilon, \delta]$ parameterization and **c, f** $[v_{pv}, v_{ph}, \delta]$ parameterization for the anisotropic Marmousi-II models

at a depth of 3 km are located in the right position compared to those in the reflection tomography models.

In Fig. 7d, ϵ inverted by the $[v_{pv}, \epsilon, \delta]$ parameterization shows long-wavelength features of the area because it is sensitive to far-offset data including large reflection angles. On

the other hand, in Fig. 7e, because the scattered wavefields of ϵ by the $[v_{ph}, \epsilon, \delta]$ parameterization are mainly affected by narrow reflection angles, the inverted ϵ shows short-wavelength features of the area focusing on the boundaries adjacent to the reservoir. With the $[v_{pv}, v_{ph}, \delta]$ parameterization,

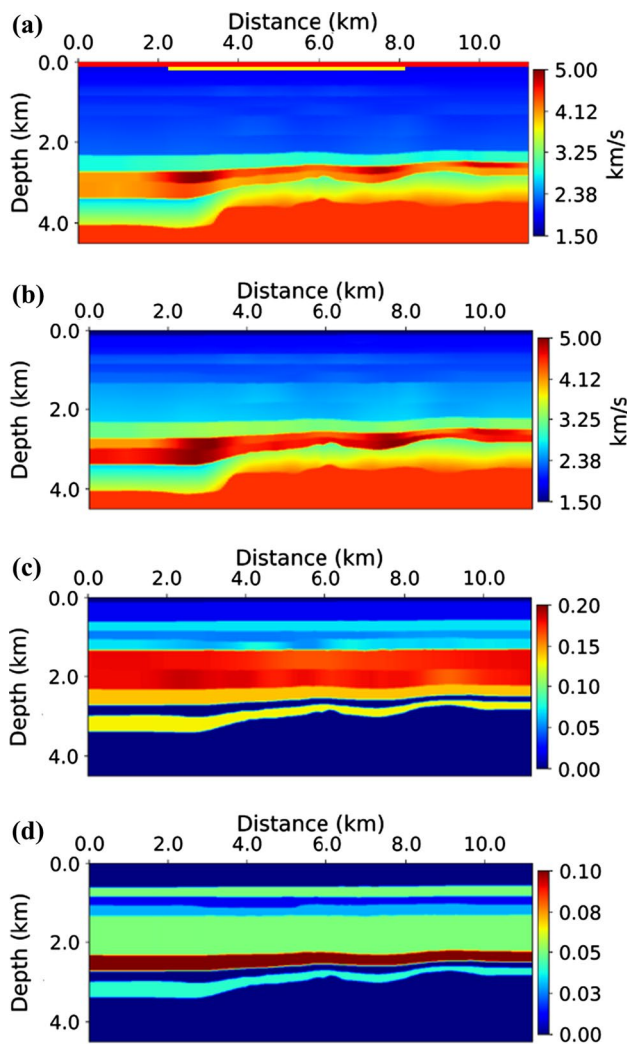


Fig. 5 2D reflection tomography models of the OBC dataset acquired in the Volve oil field: **a** v_{pv} with the source (red line) and receiver (yellow line), **b** v_{ph} , **c** ϵ , and **d** δ

ϵ is calculated indirectly from the relation between v_{pv} and v_{ph} , and the inverted v_{ph} is not properly achieved, resulting in the overestimated ϵ (Fig. 7f).

In summary, maintaining computational speed-up of the decomposed acoustic wave equation, the strategy of using the vector virtual sources provides quite reasonable multiparameter FWI results by correctly computing the scattering potentials of the anisotropic parameters depending on parameterization.

Conclusions

In multiparameter FWI, the behavior of scattered wavefields caused by the anisotropic parameter perturbations determines the dominant wavelength components according to reflection angle and the trade-off preventing model

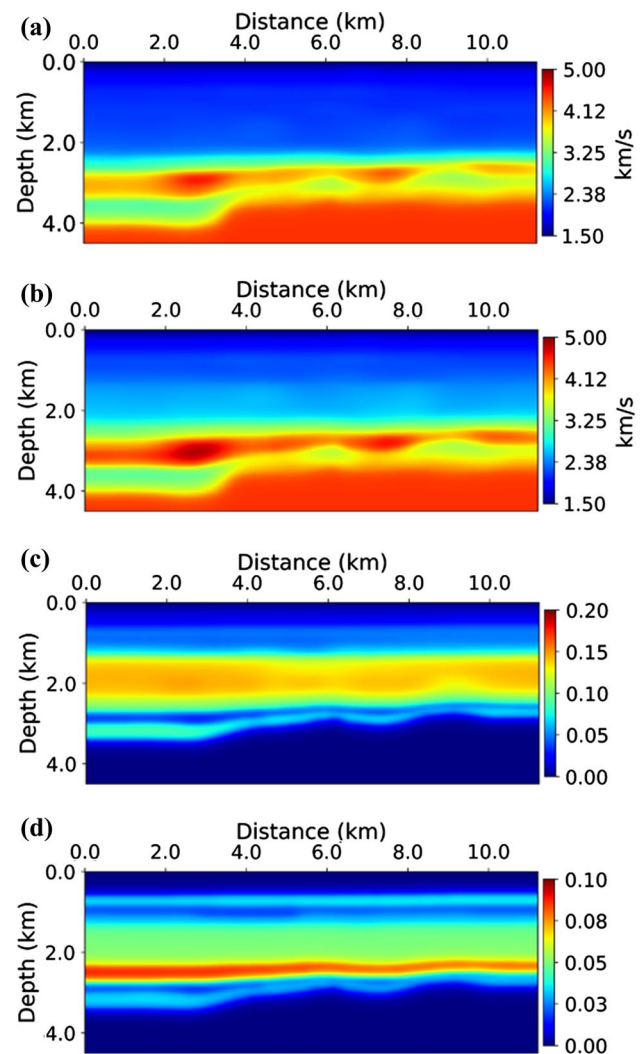


Fig. 6 Initial models used for FWI of the OBC dataset in the Volve oil field: **a** v_{pv} , **b** v_{ph} , **c** ϵ , and **d** δ

parameters being inverted independently. Thus, for multiparameter anisotropic FWI that inverts at least for two model parameters, we should investigate the scattering patterns and their consistency with the elastic scattering theory depending on parameterization. Among the acoustic approximations of the elastic wave equations for anisotropic media, the pressure-based acoustic wave equation is computationally efficient. However, the isotropic pressure virtual source from the pressure-based acoustic wave equation generates inaccurate scattering potentials, which can lead to model parameter updates in the wrong gradient direction. We investigated whether the strategy of using the vector wavefields produces the numerical PP scattering patterns consistent with the elastic scattering theory for three parameterizations: $[v_{pv}, \epsilon, \delta]$, $[v_{ph}, \epsilon, \delta]$, and $[v_{pv}, v_{ph}, \delta]$. To demonstrate the accuracy of the corrected gradient with the vector wavefields in the pressure-based acoustic

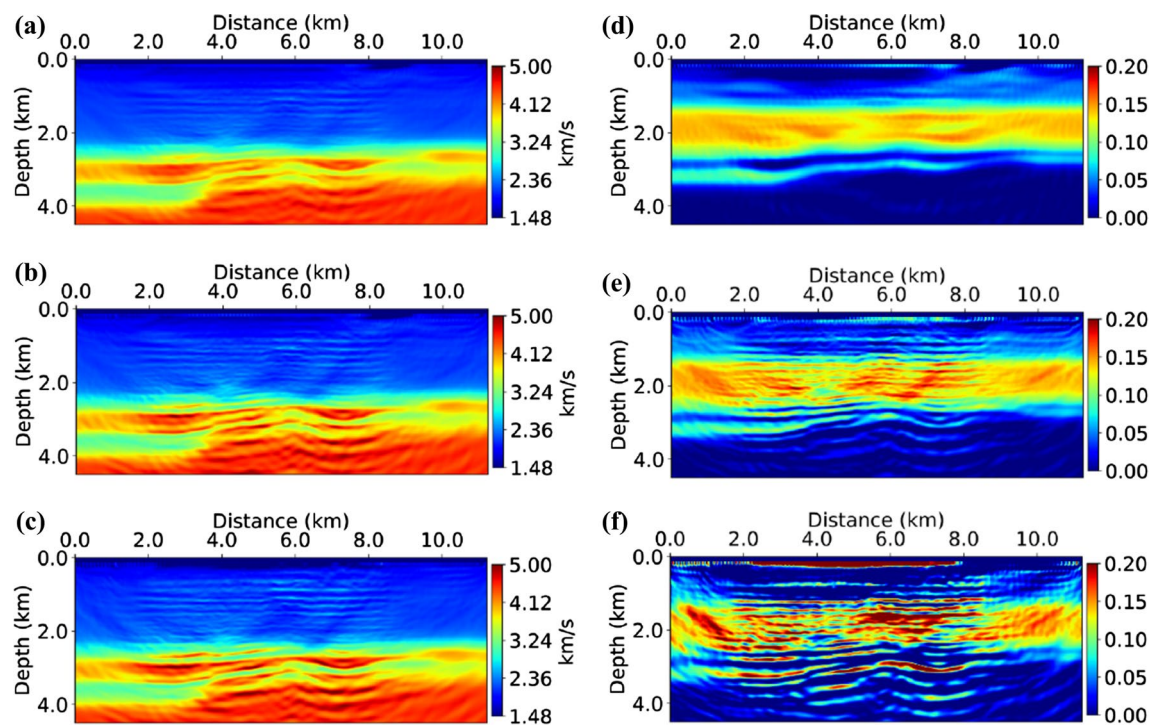


Fig. 7 FWI results of (a, b, c) v_p and (d, e, f) ϵ obtained using a, d [v_p , ϵ , δ] parameterization, b, e [v_{ph} , ϵ , δ] parameterization, and c, f [v_p , v_{ph} , δ] parameterization

FWI for three parameterizations, we conducted multiparameter acoustic FWI in 2D VTI media using the synthetic Marmousi-II data and the real OBC dataset from the North Sea. The numerical results showed that the strategy with the pressure-based acoustic wave equation provides reasonable model updates depending on the parameterization reflecting characteristics of the anisotropic parameter perturbations, while maintaining the computational speed-up from the decomposed equation as a modeling engine for FWI.

Acknowledgements This work was supported by the Basic Research Project of the Korea Institute of Geoscience and Mineral Resources (GP2020-007) funded by the Ministry of Science and ICT, by the National Foundation of Korea (NRF) funded by the Ministry of Science and ICT (NRF-2019K1A3A1A80113341, NRF-2020R111A3073977) and by the Nuclear Safety Research Program through the Korea Foundation Of Nuclear Safety (KoFONS), granted financial resource from the Nuclear Safety and Security Commission (NSSC), Republic of Korea (No.1705010). Also, the North Sea data are released by Equinor and former Volve license Partners under Creative Commons License. We greatly appreciate their efforts to disclose the Volve data. The views on the Volve data expressed in this paper are the views of the authors and do not necessarily reflect the views of Equinor and former Volve license Partners. We would like to thank the editor, an anonymous reviewer, and Prof. Tariq Alkhalifah in King Abdullah University of Science and Technology for their helpful suggestions.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

- Alkhalifah T (2000) An acoustic wave equation for anisotropic media. *Geophysics* 65:1239–1250. <https://doi.org/10.1190/1.1444815>
- Alkhalifah T (1997) An anisotropic marmousi model. SEP-95: Stanford Exploration Project:265–282
- Alkhalifah T, Tsvankin I (1995) Velocity analysis for transversely isotropic media. *Geophysics* 60:1550–1566. <https://doi.org/10.1190/1.1443888>
- Alkhalifah T (2016) Research note: Insights into the data dependency on anisotropy: An inversion prospective. *Geophys Prospect* 64:505–513. <https://doi.org/10.1111/1365-2478.12345>
- Červený V (2005) *Seismic ray theory*. Cambridge University Press, Cambridge
- Duveneck E, Milcik P, Bakker PM, Perkins C (2008) Acoustic VTI wave equations and their application for anisotropic reverse-time migration. SEG Tech Program Expanded Abstr. <https://doi.org/10.1190/1.3059320>
- Gholami Y, Brossier R, Operto S, Ribodetti A, Virieux J (2013a) Which parameterization is suitable for acoustic vertical transverse isotropic media full waveform inversion? Part 1: Sensitivity and trade-off analysis. *Geophysics* 78:R81–R105. <https://doi.org/10.1190/geo2012-0204.1>

- Gholami Y, Brossier R, Operto S, Ribodetti A, Virieux J (2013b) Which parameterization is suitable for acoustic vertical transverse isotropic full waveform inversion? Part 2: Synthetic and real data case studies from Valhall. *Geophysics* 78:R107–R124. <https://doi.org/10.1190/geo2012-0203.1>
- Kamath N, Tsvankin I, Díaz E (2017) Elastic full-waveform inversion for VTI media: a synthetic parameterization study. *Geophysics* 82:C163–C174. <https://doi.org/10.1190/geo2016-0375.1>
- Komatitsch D, Martin R (2007) An unsplit convolutional perfectly matched layer improved at grazing incidence for the seismic wave equation. *Geophysics* 72:SM155–SM167. <https://doi.org/10.1190/1.2757586>
- Lailly P (1983) As a sequence of before stack migrations. *Conference on inverse scattering. Theory and Application* 11: 206
- Oh JW, Alkhalifah T (2016) Elastic orthorhombic anisotropic parameter inversion: an analysis of parameterization. *Geophysics* 81:C279–C293. <https://doi.org/10.1190/geo2015-0656.1>
- Oh JW, Alkhalifah T (2018) Optimal full-waveform inversion strategy for marine data in azimuthally rotated elastic orthorhombic media. *Geophysics* 83:R307–R320. <https://doi.org/10.1190/geo2017-0762.1>
- Oh JW, Shin Y, Alkhalifah T, Min DJ (2020) Multistage elastic full-waveform inversion for tilted transverse isotropic media. *Geophys J Int* 223:57–76. <https://doi.org/10.1093/gji/ggaa295>
- Plessix RÉ (2006) A review of the adjoint-state method for computing the gradient of a functional with geophysical applications. *Geophys J Int* 167:495–503. <https://doi.org/10.1111/j.1365-246X.2006.02978.x>
- Plessix RÉ, Cao Q (2011) A parametrization study for surface seismic full waveform inversion in an acoustic vertical transversely isotropic medium. *Geophys J Int* 185:539–556. <https://doi.org/10.1111/j.1365-246X.2011.04957.x>
- Pratt RG, Shin C, Hicks GJ (1998) Gauss-Newton and full Newton methods in frequency domain seismic waveform inversion. *Geophys J Int* 133:341–362. <https://doi.org/10.1046/j.1365-246X.1998.00498.x>
- Shin Y, Oh JW, Kim S, Min DJ (2019) Mono-component multiparameter acoustic full waveform inversion in vertically transverse isotropic media using converted vector wavefields. *J Appl Geophys* 170:103816. <https://doi.org/10.1016/j.jappgeo.2019.07.010>
- Tarantola A (1984) Inversion of seismic reflection data in the acoustic approximation. *Geophysics* 49:1259–1266. <https://doi.org/10.1190/1.1441754>
- Thomsen L (1986) Weak elastic anisotropy. *Geophysics* 51:1954–1966. <https://doi.org/10.1190/1.1442051>
- Virieux J, Operto S (2009) An overview of full-waveform inversion in exploration geophysics. *Geophysics* 74:WCC1–WCC26. <https://doi.org/10.1190/1.3238367>
- Wu Z, Liu H, Alkhalifah T (2018) Pure Quasi-P-wave calculation in transversely isotropic media using a hybrid method. *Geophys J Int* 214:421–429. <https://doi.org/10.1093/gji/ggy151>
- Xu S, Zhou H (2014) Accurate simulations of pure quasi p-waves in complex anisotropic media. *Geophysics* 79:T341–T348. <https://doi.org/10.1190/geo2014-0242.1>
- Xu S, Tang B, Mu J, Zhou H (2015) Elliptic decomposition of quasi-P wave equation. *EAGE Tech Program Ext Abstr*. <https://doi.org/10.3997/2214-4609.201413134>
- Zhou H, Zhang G, Bloor R (2006) An anisotropic acoustic wave equation for VTI media. *EAGE Tech Program Ext Abstr*. <https://doi.org/10.3997/2214-4609.201402310>